

HORN ISLAND, AustraliaWMO: **941740**Lat: **10.5844S**Lon: **142.2900E**Elev: **16**StdP: **14.69**Time Zone: **10.00 (E10)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	71.2	72.5	61.6	81.9	76.9	63.1	86.4	76.8	25.3	81.9	24.2	82.1	6.1	150	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.1	89.9	78.8	89.2	78.7	88.1	78.5	81.9	85.5	80.9	85.5	80.3	85.1	14.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.1	83.4	79.9	155.3	83.1	79.0	150.7	82.9	45.8	85.7	44.7	85.6	44.0	85.5	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 23.3	22.1	20.8		66.3	92.0	3.0	1.0	64.1	92.7	62.4	93.3	60.8	93.9	58.6	94.6	
(5)				63.7	83.5	2.4	2.2	61.9	85.0	60.5	86.3	59.2	87.6	57.4	89.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.2	82.2	81.9	81.8	82.0	81.1	79.5	78.5	78.4	79.9	81.6	83.3	83.8	(6)
(7)		DBStd	2.44	1.96	1.85	1.78	1.64	1.60	1.58	1.64	1.87	1.81	1.88	1.61	1.94	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11371	998	893	986	960	965	886	882	880	897	979	998	1047	(10)
(11)		CDD65	5896	533	473	521	510	500	436	417	415	447	514	548	582	(11)
(12)		CDH74	59065	6048	5272	5638	5510	4872	3540	2897	2884	3761	5198	6337	7108	(12)
(13)		CDH80	16416	1966	1594	1627	1549	1105	559	371	405	748	1420	2239	2833	(13)
(14)	Wind	WSAvg	12.1	8.6	7.9	7.4	10.7	13.5	14.0	14.8	15.7	15.0	12.9	9.3	(14)	
(15)	Precipitation	PrecAvg	69.1	16.9	15.0	14.5	8.2	1.5	0.8	0.4	0.2	0.3	1.3	9.6	(15)	
(16)		PrecMax	102.9	34.0	27.8	25.6	28.2	5.7	4.8	1.3	0.6	0.6	1.5	7.3	31.3	(16)
(17)		PrecMin	35.9	6.7	5.0	3.5	0.6	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	(17)
(18)		PrecStd	15.2	6.9	5.9	5.7	6.6	1.5	1.0	0.3	0.2	0.2	0.4	1.8	7.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	89.6	89.7	88.5	87.7	86.2	85.2	86.1	87.5	89.3	90.7	91.5	(19)
MCWB			79.3	80.1	80.1	79.2	78.6	76.3	75.7	76.0	77.2	77.6	79.5	79.0	(20)	
2%		DB	88.8	88.1	88.1	87.8	86.3	84.8	84.1	84.3	85.9	87.9	89.3	89.9	(21)	
		MCWB	79.1	79.5	79.7	79.3	77.8	75.8	74.5	74.3	75.7	77.3	78.3	78.5	(22)	
5%		DB	87.8	87.4	87.3	86.8	85.8	84.1	82.9	83.1	84.6	86.7	88.1	89.4	(23)	
		MCWB	79.1	79.3	79.4	78.6	77.4	75.4	73.6	73.2	74.3	76.5	77.8	78.4	(24)	
10%		DB	86.5	86.2	86.1	86.0	84.6	82.9	82.3	82.3	83.9	85.9	87.5	88.2	(25)	
		MCWB	78.9	79.0	78.8	78.1	76.4	74.7	73.3	72.6	73.8	76.0	77.5	78.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.9	83.2	83.3	82.7	83.2	79.2	78.3	77.7	78.3	79.9	82.3	81.0	(27)
MCD B			86.2	85.1	86.5	85.4	83.7	82.4	81.3	82.6	84.8	84.9	86.3	87.5	(28)	
2%		WB	80.8	81.5	81.0	80.8	79.5	77.6	76.5	76.2	77.2	78.9	80.7	80.1	(29)	
		MCD B	85.7	84.9	85.5	85.0	83.7	82.3	80.8	81.5	83.0	84.5	86.1	86.5	(30)	
5%		WB	80.1	80.5	80.1	80.0	78.5	76.6	75.7	75.3	76.2	78.1	79.6	79.7	(31)	
		MCD B	85.0	84.8	84.6	84.2	83.4	81.6	80.4	80.6	82.2	84.0	85.4	86.0	(32)	
10%		WB	79.5	79.8	79.6	79.2	77.8	76.0	74.8	74.3	75.3	77.3	78.8	79.2	(33)	
		MCD B	84.3	84.1	83.9	83.4	82.8	81.0	79.9	80.0	81.7	83.5	84.8	85.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	7.9	8.1	7.8	7.9	8.1	8.4	8.9	8.9	9.1	9.0	9.1	(35)
MCDBR			9.6	9.4	9.5	8.7	8.4	8.5	8.7	9.2	9.1	9.6	9.6	10.0	(36)	
MCWBR			4.0	4.1	4.1	3.9	3.8	3.4	3.3	3.6	3.4	3.6	3.7	3.8	(37)	
5% WB		MCDBR	8.7	8.5	8.7	8.0	7.7	7.9	7.8	8.7	8.8	9.1	9.0	9.3	(38)	
		MCWBR	4.4	4.6	4.4	4.3	4.0	3.5	3.4	3.7	3.3	3.6	3.7	4.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.408	0.406	0.397	0.403	0.410	0.404	0.396	0.400	0.411	0.415	0.417	0.410	(40)	
taud		2.559	2.564	2.594	2.564	2.520	2.531	2.545	2.522	2.480	2.464	2.475	2.524	(41)		
Ebn at Noon		296	295	293	281	270	266	272	279	284	289	292	295	(42)		
Edh at Noon		34	34	32	32	32	30	30	33	36	37	37	36	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1562	1615	1511	1534	1522	1412	1494	1727	1912	2072	2052	1858	(44)	
(45)		RadStd	179	197	102	176	112	90	84	129	121	102	157	233	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page